

Dynamic Scene Adjustment for Player Engagement in VR Game

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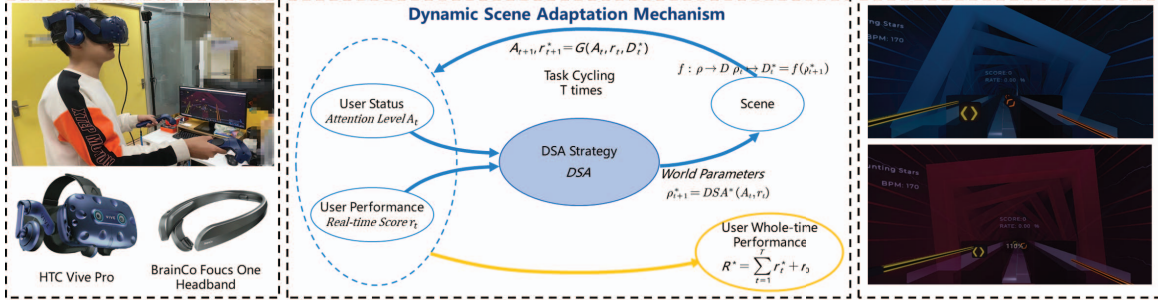


Figure 1: The left part illustrates participants wearing VR and EEG Headband equipment to conduct experiments; the right part is the game scenes before and after DSA strategy adjustment. The middle part is the DSA Mechanism pipeline, which details will be introduced in the second section.

ABSTRACT

Virtual reality (VR) produces a highly realistic simulated environment with controllable environment variables. This paper proposes a Dynamic Scene Adjustment (DSA) mechanism based on the user interaction status and performance, which aims to adjust the VR experiment variables to improve the user's game engagement. We combined the DSA mechanism with a musical rhythm VR game. The experimental results show that the DSA mechanism can improve the user's game engagement (task performance).

Index Terms: Human-centered computing—Interaction paradigms—Virtual reality;

1 INTRODUCTION

The environment can have a significant impact on user performance and experience. Virtual reality (VR) produces a highly realistic simulated environment with controllable environment variables. Many studies have proved that user status, experience, or performance can be affected by variables such as color, music, and light intensity in the environment [2]. However, few researchers have tried to dynamically adjust the user's state in real-time by controlling these VR environmental variables.

This work aims to establish a Dynamic Scene Adjustment (DSA) mechanism based on the user interaction state, as illustrated in Fig. 1. To enhance the user experience and overall performance (the core of user engagement), the DSA mechanism dynamically modifies the VR environment variables according to the user's instant status

and performance. The inspiration for this research comes from two aspects: a) When we physiologically monitor users' attention in the VR environment, we found that some experiment variables like color significantly impact the user's attention and performance during the task, which has also been reported in previous research [3, 5]. b) Referring to the mature Difficulty Dynamic Adjustment method [6], it should also be reasonable to improve the users' experience and overall performance by dynamically adjusting the VR environment through adaptive user's instant status and performance.

We also designed a DSA game demo based on the open-source VR game Beat Saber¹. We invited 13 participants to conduct a total of 18 groups of comparative experiments on the DSA demo game. The experimental results show that adjusting the color (experiment variable) in the VR environment through adaptive user attention (status) and instant performance can simultaneously improve their experience and overall performance. This also proves the feasibility of the DSA mechanism. The user's engagement (experience, status, and overall performance) during the interaction process can be optimized by adjusting the interaction environment parameters.

2 DYNAMIC SCENE ADJUSTMENT MECHANISM

Design of the Dynamic Scene Adjustment (DSA) mechanism. Users are not only unilaterally affected by the environment; the decisions made at every moment and their status, performance, and experience can also affect the user's state at the next moment. Thus, the DSA mechanism can be described with the following equations.

$$\begin{aligned} \rho_{t+1}^* &= DSA^*(A_t, r_t) & f: \rho \rightarrow D, \rho_t \mapsto D_t^* &= f(\rho_{t+1}^*) \\ A_{t+1}, r_{t+1}^* &= G(A_t, r_t, D_t^*) & R^* &= \sum_{t=1}^T r_t^* + r_0 \end{aligned} \quad (1)$$

where ρ is the world parameters, the function $DSA^*(\cdot)$ is the best DSA intervention strategy, A_t is users' status, r_t is users' current performance, r_{t+1}^* is the user's performance after enhanced by the DSA mechanism at the next moment, A_{t+1} is the status when the

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¹<https://beatsaber.com/>

user interacts with the scene adjusted by the DSA mechanism, the function $G()$ is the game-play rules. R^* is the best user overall interaction performance, which is also the adjustment goal of the DSA mechanism. D is the user's action, which satisfies the mapping relationship f with ρ . The intervention strategy DSA calculates the world parameters with users' status and performance as input. Many intervention strategies exist, and the best strategy DSA^* will obtain the most suitable parameter ρ^* . ρ^* will cause the user to take the best action D^* . After the user's state and action interact with the gameplay, their new status A_t and performance r_{t+1} will occur. This is the logic of the operation of the DSA mechanism. Therefore, the DSA mechanism mainly includes three modules. a) **World parameters** are the environmental variables that mainly affect the user's performance and experience. b) **Adjustment goal** is the expected state or desired experience of the users (or the anticipated state the examiner hopes participants can complete). c) **Intervention strategy** is the calculation function of world parameters, which is the logic about how users achieve the adjustment goal.

Gameplay design with DSA mechanism. In this part, we introduce our DSA game to readers according to the three main modules required by DSA, as illustrated in Fig. 2. We selected the background color in the VR environment as the world parameter ρ and set it to either red or blue. Because the selected game is a kind of musical rhythm VR game, it has been proven that color can influence users' attention, Red can stimulate users' attention, and Blue will keep users' attention stable [5]. Users' attention will influence users' experience [1] and performance [4]. We set the users' game score (user performance) in the VR environment as the adjustment goal. Regarding the intervention strategy, we input the user's real-time attention and instant performance into the DSA strategy module, dynamically adjusting the world parameters selected (color) in this game. Users' instant performance is calculated by the user's score changes within an adjacent time window. The adjustment strategies of the game are illustrated in Table 1. We assume that the user's score ratio in the first time window is S_t , while in the second time window is S_{t+1} . Users' instant performance $r_{t+1} = S_{t+1} - S_t$. The time window size is 2.5 seconds which is the hyper-parameters obtained after decoupling the game.

3 EXPERIMENT SETUP AND RESULTS ANALYSIS

Experiment Setup The experiment was supervised and guided by the Ethics Committee of UESTC. 13 participants, including 4 female and 9 male students between 20 and 28 years old, completed the comparative experiment. The game with (or without) the DSA mechanism is used as the experimental (control) group. Participants were asked to be involved twice in each experimental and control group experiment in random order. To avoid the impact of proficiency, all participants must practice several times before the formal experiment. Participants are given enough rest time after each game to avoid the impact of fatigue. The user's attention and task performance (game score) are collected and saved during the experiments. A short user experience interview is conducted after the experiment.

Experiment Results With the DSA mechanism, 83.3% of the participants improved their performance. The paired sample t-test results show that the test subjects performed better in the experimental group ($M=86.13278$, $SD=8.79289$) than in the control group ($M=84.47861$, $SD=9.7518$). The experimental results show that the **DSA mechanism can improve the user's task performance in the VR environment**. In addition, according to interview feedback, almost all participants reported that the experimental group was more excited and they experienced better than the control group.

4 CONCLUSION

In summary, this paper proposes the Dynamic Scene Adjustment (DSA) mechanism for dynamically improved user interaction performance by adjusting the world parameters in the VR environment

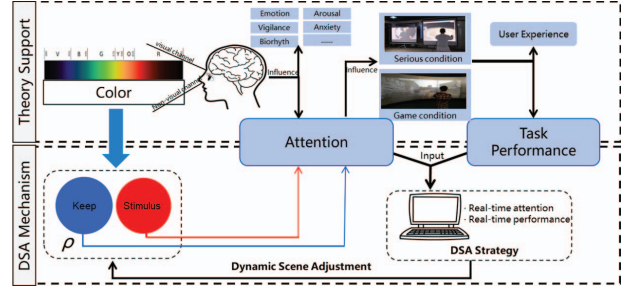


Figure 2: DSA game pipeline, the upper part of the figure is the theoretical basis for adjustment goals and intervention strategies. The lower part is the specific DSA logic applied in the game

Table 1: Dynamic Scene Adjustment Strategy

No.	User Performance (Score Ratio)	User Status (Attention Level)	World Parameters (Color)
1	$S_{t+1} - S_t \geq 0$	High	Blue
2	$S_{t+1} - S_t \geq 0$	Low	Red
3	$S_{t+1} - S_t < 0$	Low	Red
4	$S_{t+1} - S_t < 0$	High	Maintain current scene color

following the user's real-time status and instant performance. In addition, we combined the DSA mechanism with a musical rhythm VR game and applied it to VR environment tasks. The experimental conclusions support the rationality of the mechanism. This work can help researchers think about dynamic regulation in VR environments from a new perspective. It can inspire the design of VR therapy, VR education, VR games, and other fields. Future research will expand the experimental data based on the existing DSA mechanism, adding more world parameters. In addition, we will want to establish a universal DSA mechanism through a data-driven approach.

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